Quizrr Chapter-Wise Test for JEE Advanced - 2025

By: C I P H Ξ R

Q.1

Choose the correct statements -
(1) Designation for the orbital with the quantum number $\mathrm{n}=3, \ell=1, \mathrm{~m}=-1$ may be $3 \mathrm{p}_{\mathrm{x}}$ or $3 \mathrm{p}_{\mathrm{y}}$
(2) Designation for the orbital with the quantum number $\mathrm{n}=4, \ell=2, \mathrm{~m}=+2$ may be $4 \mathrm{~d}_{\mathrm{xy}}$ or $4 \mathrm{~d}_{\mathrm{x}^2-\mathrm{y}^2}$
(3) Designation for the orbital with the quantum number $\mathrm{n}=5, \ell=0, \mathrm{~m}=0$ may be $5 \mathrm{~s}$
(4) Designation for the orbital with the quantum number $\mathrm{n}=2, \ell=1, \mathrm{~m}=0$ may be $3 \mathrm{p}_{\mathrm{z}}$

(A) 1,2 and 3 are correct

(B) 1 and 2 are correct

(C) 2 and 4 are correct

(D) 1 and 3 are correct

Q.2

Choose the correct options related to no. of electrons in a atom with the following quantum numbers -
(1) $\mathrm{n}=4, \ell=1 ; 4$
(2) $\mathrm{n}=2, \ell=1, \mathrm{~m}=-1, \mathrm{~s}=+1/2 ; 1$
(3) $\mathrm{n}=3 ; 9$ electrons
(4) $\mathrm{n}=4, \ell=2, \mathrm{~m}=0 ; 2$

(A) 1,2 and 3 are correct

(B) 1 and 2 are correct

(C) 2 and 4 are correct

(D) 1 and 3 are correct

Q.3

Choose the correct options -
(1) In case of half filled and completely filled orbitals, the exchange energy is maximum and is greater then the loss of orbital energy due to the transfer of electron from a higher to a lower sublevel
(2) The greater the number of possible exchanges between the electrons of parallel spins present in the degenerate orbitals, the higher would be the amount of energy released and more will be the stability.
(3) Binding energy is the measure of stability of the nucleus.
(4) If the value of binding energy is negative then the stability order is Product nucleus $>$ reactant nucleus

(A) 1,2 and 3 are correct

(B) 1 and 2 are correct

(C) 2 and 4 are correct

(D) 1 and 3 are correct

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Q.4

The isoelectronic pair of 32 electron is

(A) BO_3^{-3} and CO_3^{-2}

(B) PO_4^{-3} and CO_3^{-2}

(C) N_2 and CO

(D) All of the above

Q.5

An electron in a hydrogen atom in its ground state absrobs energy equal to the ionisation energy of Li^{+2}. The wavelength of the emitted electron is

(A) 3.32×10^{-10} m

(B) $1.17\mathring{A}$

(C) 2.32×10^{-9} nm

(D) 3.33pm

Q.6

Assuming Heisenberg Uncertainity Principle to be true what could be the miniumum uncertainity in de-Broglie wavelength of a moving electron accelerated by Potential Difference of 6 V whose uncertainity in position is $7/22$n.m.

(A) $6.25\mathring{A}$

(B) $6\mathring{A}$

(C) $0.625\mathring{A}$

(D) $0.3125\mathring{A}$

Q.7

The pair $NH_3 + BH_3$ is isoelectronic with

(A) B_2H_6

(B) C_2H_6

(C) C_2H_4

(D) CO_2

Q.8

An electron collides with a hydrogen atom in its ground state and excited it to a state of $n = 3$. How much energy was given to the hydrogen atom in this inelastic collision -

(A) 12.08eV

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(B) 6.12eV

(C) 15.14eV

(D) 18.21eV

Q.9

The radial distribution functions $[P(r)]$ is used to determine the most probable radius, which is used to find the electron in a given orbital $\frac{\mathrm{dP(r)}}{\mathrm{dr}}$ for 1 s-orbital of hydrest number Z, is $\frac{\mathrm{dP}}{\mathrm{dr}} = \frac{4Z^3}{\mathrm{a_0^3}}\left(2\mathrm{r} - \frac{2Z\mathrm{r}^2}{\mathrm{a_0}}\right)\mathrm{e}^{-2\mathrm{Zr}/\mathrm{a_0}}$ then which of the following statements is/are correct?

(A) At the point of maximum value of radial distribution function $\frac{\mathrm{dP(r)}}{\mathrm{dr}} = 0$, one antinode is present.

(B) Most probable radius of Li^{2+} is $\frac{\mathrm{a_0}}{3}\mathrm{pm}$.

(C) Most probable radius of $\mathrm{He}^{+}$is $\frac{\mathrm{a_0}}{2}\mathrm{pm}$.

(D) Most probable radius of hydrogen atom is $\mathrm{a_0pm}$.

Q.10

In a sample of H-atom for $5 \rightarrow 2$ transition. Choose the correct options -

(A) Maximum 6 spectral lines will be observed from the sample

(B) A photon of wavelength $\lambda = \frac{36}{5\mathrm{R}}$ can be emitted

(C) A photon of frequency $\bar{v} = \frac{7R}{144}$ can be emitted.

(D) When single isolated atom/ion is considered, maximum 4 spectral lines are observed for given transition.

Q.11

In a sample of hydrogen atom like ion. electron is in 2nd excited state. the binding energy for electron in fourth orbit of this atom is 13.6eV then correct statements is/are

(A) a 25eV photon can set electron free from second excited state of this sample

(B) 3 different types of photon will be observed if electrons make transition up to ground state from the second excited state

(C) Fourth line of Balmer series for this hydrogen like ion has same energy as of 2nd line of Lyman series in He^{+} ion spectrum

(D) Energy gap for this H-like ion between 3rd and 4th orbits is equal to gap between Ist and 2nd orbits of H-atom

Q.12

Passage:

Werner Heisenberg considered the limits of how precisely we can measure the properties of an electron or other microscopic particle. He determined that there is a fundamental limit to how closely we can measure both position and momentum. The more accurately we measure the momentum of a particle, the less accurately we can determine its position. The converse is also true. This is summed up in what we now call the Heisenberg uncertainty principle. The equation is Δ x. $\Delta(\mathrm{mv}) \geq \frac{\mathrm{h}}{4\pi}$.

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The uncertainty in the position or in the momentum of a macroscopic object like a baseball is too small to observe. However, the mass of microscopic object such as an electron is small enough for the uncertainty to be relatively large and significant.

Question:

If the uncertainties in position and momentum are equal, the uncertainty in the velocity is -

(A) $\sqrt{\frac{\mathrm{h}}{\pi}}$

(B) $\sqrt{\frac{\mathrm{h}}{2\pi}}$

(C) $\frac{1}{2m}\sqrt{\frac{h}{\pi}}$

(D) None of these

Q.13

Passage:

Werner Heisenberg considered the limits of how precisely we can measure the properties of an electron or other microscopic particle. He determined that there is a fundamental limit to how closely we can measure both position and momentum. The more accurately we measure the momentum of a particle, the less accurately we can determine its position. The converse is also true. This is summed up in what we now call the Heisenberg uncertainty principle. The equation is Δ x. $\Delta(\mathrm{mv}) \geq \frac{\mathrm{h}}{4\pi}$.

The uncertainty in the position or in the momentum of a macroscopic object like a baseball is too small to observe. However, the mass of microscopic object such as an electron is small enough for the uncertainty to be relatively large and significant.

Question:

If the uncertainty in velocity and position is same, then the uncertainty in momentum will be -

(A) $\sqrt{\frac{\mathrm{hm}}{4\pi}}$

(B) $m\sqrt{\frac{h}{4\pi}}$

(C) $\sqrt{\frac{\mathrm{h}}{4\pi\mathrm{m}}}$

(D) $\frac{1}{m}\sqrt{\frac{h}{4\pi}}$

Q.14

Passage:

Werner Heisenberg considered the limits of how precisely we can measure the properties of an electron or other microscopic particle. He determined that there is a fundamental limit to how closely we can measure both position and momentum. The more accurately we measure the momentum of a particle, the less accurately we can determine its position. The converse is also true. This is summed up in what we now call the Heisenberg uncertainty principle. The equation is Δ x. $\Delta(\mathrm{mv}) \geq \frac{\mathrm{h}}{4\pi}$.

The uncertainty in the position or in the momentum of a macroscopic object like a baseball is too small to observe. However, the mass of microscopic object such as an electron is small enough for the uncertainty to be relatively large and significant.

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Question:

What would be the minimum uncertainty in de-Broglie wavelength of a moving electron accelerated by 6 volt

(A) $6.25\mathring{A}$

(B) $6\mathring{A}$

(C) $0.625\mathring{A}$

(D) $0.3125\mathring{A}$

Q.15

Passage:

Einstein had suggested that light can behave as a wave as well as like a particle i.e. it has dual character. In 1924 , de-Broglie proposed that an electron, behaves both as a material particle and as a wave. This proposed a new theory wave mechanical theory of matter. According to this theory, the electrons protons and even atom when in motion possess wave properties. According to de-Broglie, the wavelength associated with a particle of mass m, moving with velocity v is given by the relation, $\lambda = \frac{\mathrm{h}}{\mathrm{mv}}$, where h is Planck's constant.

Question:

$\mathrm{K.E.}$ of the electron is 4.55×10^{-25} J. Its de Broglie wave length is -

(A) $4700\mathring{A}$

(B) $8300\mathring{A}$

(C) $7200\mathring{A}$

(D) $7400\mathring{A}$

Q.16

Passage:

Einstein had suggested that light can behave as a wave as well as like a particle i.e. it has dual character. In 1924 , de-Broglie proposed that an electron, behaves both as a material particle and as a wave. This proposed a new theory wave mechanical theory of matter. According to this theory, the electrons protons and even atom when in motion possess wave properties. According to de-Broglie, the wavelength associated with a particle of mass m, moving with velocity v is given by the relation, $\lambda = \frac{\mathrm{h}}{\mathrm{mv}}$, where h is Planck's constant.

Question:

For particles having same kinetic energy, the de Broglie wavelength is -

(A) Directly proportional to its velocity

(B) Inversely proportional to its velocity

(C) Independent of velocity and mass

(D) Unpredictable.

Q.17

Wavelength of photon emited when an electron in H-atom jumps from an orbital having 2 angular nodes with one

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spherical node to an orbital having total 1 node is -

Given $hc = 1240\text{eV.nm}$

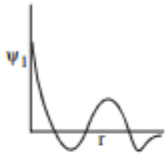
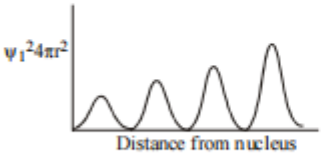
(A) $\frac{1240}{10.2}$ nm

(B) $\frac{1240}{1.89}$ nm

(C) $\frac{1240}{2.55}$ nm

(D) $\frac{1240}{0.66}$ nm

Q.18

Column I	Column II
(A) 	(p) 4s
(B) 	(q) $5p_y$
(C) Angular probability is dependent of θ and ϕ.	(r) 3s
(D) Atleast one angular node is present	(s) $6d_{xy}$

(A) A-p; B-p,q,s ; C- q,s ; D-q,s;

(B) A-p,q; B-r ; C- p,q,s ; D-p,q,s;

(C) A-p,q,s; B-p,r ; C- p,q,r,s ; D-p,q,s;

(D) A-r,s; B-p,r ; C- p ; D-p,q,s;

Q.19

Suppose that at time t the state function of a one-particle system is:

$\Psi = \left(2/\pi C^2\right)^{3/4} e^{-\left(x^2+y^2+z^2\right)/C^2}$ where $C = 2$ nm.

The probability of finding the particle at any time in the tiny cubic region with it's centre at $x = 1.2\,\text{nm},\ \text{y} = -1.0\,\text{nm}$ and $\text{z} = 0$ and with edges of each of length $0.004\,\text{nm}$ is 1.15×10^{-K} (In scientific notation). Find K

$\left\{\text{e}^{-1.22} = 0.30\right), \left(\frac{2}{4\pi}\right)^{3/2} = 0.06\}$

Q.20

Photochemical dissociation of oxygen results in the production of two oxygen atoms, one in the ground state and one in the excited state.

$\text{O}_2 \xrightarrow{\text{hv}} \text{O} + \text{O}^*$

The maximum wavelength (λ) needed for this is $174\,\text{nm}$. If the excitation energy $\text{O} \rightarrow \text{O}^*$ is 3.15×10^{-19} J. $49.83 \times 10^{\text{x}}\text{kJ/mol}$ energy is needed for the dissociation of one mole of oxygen into normal atoms in ground state. Find the value of x.

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Answers & Solutions

Q.1 Answer:

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Solution:

(a) Since $\ell = 1$ corresponds to p-orbital and $m = -1$ shows orientation either in x or y axis, thus this orbital refers to $3p_x$ or $3p_y$
(b) $4\ d_{xy}$ or $4\ d_{x^2-y^2}$
(c) $5\ s$
(d) $2p_z$

Q.2 Answer:

2 and 4 are correct

Solution:

(a) $\ell = 1$ refers to p - subshell which has three orbitals $(p_x, p_y$ and $p_z)$ each having two electrons. Therefore, total number of electrons are 6 .
(b) $\ell = 1$ refers to p - subshell, $m = -1$ refers to p_x or p_y orbital whereas, $s = +1/2$ indicate for only 1 electron.
(c) For $n = 3, I = 0, 1, 2$ $\ell = 0$ $m = 0$ 2 electrons $\ell = 1$ $m = -1$ 6 electrons $\ell = 2$ $m = -2, -1, 0, +1, +2$ 10 electrons
Total electrons 18 electrons
Alternatively, number of electrons for any energy level is given by $2n^2$ i.e. $2 \times 3^2 = 18$ electrons.
(d) $\ell = 2$ means d-subshell and $m = 0$ refer to d_{z^2} orbital
$\therefore$ Number of electrons are 2.

Q.3 Answer:

1 and 2 are correct

Solution:

(a) Binding energy per nucleon is the measure of stability of the nucleus.
(b) If the value of binding energy is negative then the stability order is
Product nucleus $<$ reactant nucleus

Q.4 Answer:

BO_3^{-3} and CO_3^{-2}

Solution:

$BO_3{}^{-3}$	$CO_3{}^{-2}$
$5 + 24 + 3 = 32$	$6 + 24 + 2 = 32$

Q.5 Answer:

$1.17\mathring{A}$

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Solution:

$$\frac{1}{\lambda} = 109677 \times 9$$
$$\lambda = 1.01 \times 10^{-6}\ \text{cm}$$
$$E = \frac{6.62 \times 10^{-34} \times 3 \times 10^{8}}{1.01 \times 10^{-8}}$$
$$= 19.66 \times 10^{-18}\ \text{J}$$
$$\Rightarrow \frac{1}{2}\text{mv}^2 = 19.66 \times 10^{-18}$$
$$\text{V}^2 = \frac{39.32 \times 10^{-18}}{9.1 \times 10^{-31}}$$
$$\text{V}^2 = 4.32 \times 10^{13}$$
$$V^2 = 43.2 \times 10^{12}$$
$$V = 6.57 \times 10^{6}$$
$$\lambda_{\text{debroglie}} = \frac{6.62 \times 10^{-34}}{9.1 \times 10^{-31} \times 6.57 \times 10^{6}}$$
$$= 1.17 \times 10^{-10}\ \text{m}$$
$$= 1.17\mathring{A}$$

Q.6 Answer:

$0.625\mathring{A}$

Solution:

$$\frac{1}{2}\text{mV}^2 = 6 \times 1.6 \times 10^{-19}$$
$$\text{V}^2 = \frac{12 \times 1.6 \times 10^{-19}}{9.1 \times 10^{-31}}$$
$$\text{V}^2 = 2.10 \times 10^{12}$$
$$\text{V} = 1.44 \times 10^{6}\ \text{m/sec}$$
$$\lambda = \frac{6.62 \times 10^{-34}}{1.44 \times 10^{6} \times 9.1 \times 10^{-31}}$$
$$= 0.5 \times 10^{-9}$$
$$\Delta x \cdot \frac{\Delta\lambda}{0.25 \times 10^{-18}} = \frac{1}{\lambda}$$
$$\frac{7}{22} \times 10^{-9} \times \frac{\Delta\lambda}{0.25 \times 10^{-18}} = \frac{1}{4\pi}$$
$$\Delta\lambda = \frac{0.25 \times 10^{-9}}{4}$$
$$= 0.0625 \times 10^{-9}$$
$$= 0.625\mathring{A}$$

Q.7 Answer:

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C_2H_6

Solution:

$NH_3 + BH_3$ C_2H_6

$7 + 3 + 5 + 3 = 18$ $6 \times 2 + 6 = 18$

Q.8 Answer:

12.08eV

Solution:

From equation $E_n = \frac{E_1}{n^2}$, the energy change of a H atom that goes from initial state of quantum number n_i to a final state of quantum number n_f is $\Delta E = E_f - E_i = \frac{E_1}{n_1^2} - \frac{E_1}{n_1^2} = E_1\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$ Hence

$n_i = 1, n_f = 3, E_i = 3, E_1 = -13.6\text{eV}$

$\Delta\text{E} = -13.6\left(\frac{1}{3^2} - \frac{1}{1^2}\right)\text{eV} = 12.08\text{eV}.$

Q.9 Answer:

At the point of maximum value of radial distribution function $\frac{\text{dP(r)}}{\text{dr}} = 0$, one antinode is present.

Solution:

At the point of maximum value of RDF

$$\frac{\text{dP}}{\text{dr}} = 0; \left(2\text{r} - \frac{2\text{Zr}^2}{\text{a}_0}\right) = 0; \text{r} = \frac{\text{a}_0}{\text{Z}}$$

where $Z = 3$ for Li^{2+} and $Z = 2$ for the He^+, $Z = 1$ for hydrogen.

Q.10 Answer:

Maximum 6 spectral lines will be observed from the sample

Solution:

For $5 \rightarrow 2$ transition in H-atom sample

Max. spectral lines $= \frac{\Delta\text{n}(\Delta\text{n}+1)}{2} = \frac{3\times4}{2} = 6$

λ of photon corresponding to $3 \rightarrow 2$ transition : $\frac{1}{\lambda} = R(1)^2 \quad \left[\frac{1}{2^2} - \frac{1}{3^2}\right] \therefore \lambda = \frac{36}{5R}$

of photon corresponding to $4 \rightarrow 3$ transition : $\bar{v} = R(1)^2 \quad \left[\frac{1}{3^2} - \frac{1}{4^2}\right] \therefore \bar{v} = \frac{7R}{144}$

For single isolated atom, maximum spectral lines $= 3 \quad (5 \rightarrow 4, 4 \rightarrow 3, 3 \rightarrow 2)$

Q.11 Answer:

a 25eV photon can set electron free from second excited state of this sample

Solution:

BE of 4th state $= 13.6 \times \frac{z^2}{n^2} = 13.6$

$\therefore z = 4$

Energy of e^- in 3rd state $= 13.6 \times \frac{16}{9} = 25\text{ev}.$

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For 'C' option

$$\Delta E_1 = 13.6 \times 16\left[\frac{1}{4} - \frac{1}{16}\right] = 40.8\text{ev}.$$

$$\Delta E_2 = 13.6 \times 4\left[\frac{1}{1} - \frac{1}{9}\right] = 40.35\text{ev}.$$

For 'D' option

$$\Delta E_1 = 13.6 \times 16\left[\frac{1}{9} - \frac{1}{16}\right] = 10.57\text{ev}$$

$$\Delta E_2 = 13.6\left[\frac{1}{1} - \frac{1}{4}\right] = 10.2\text{ev}$$

Q.12 Answer:

$\frac{1}{2m}\sqrt{\frac{h}{\pi}}$

Solution:

(i) $\Delta x \Delta p = \frac{h}{4\pi} \Rightarrow \Delta p^2 = \frac{h}{4\pi}$

$$m^2\Delta v^2 = \frac{h}{4\pi} \Rightarrow \Delta v = \frac{1}{2\,m}\sqrt{\frac{h}{\pi}}$$

(ii) $\Delta x = \sqrt{\frac{h}{4\pi m}}, \Delta x \Delta p = \frac{h}{4\pi}$

$$\sqrt{\frac{h}{4\pi m}}\Delta p = \frac{h}{4\pi}; \Delta p = \sqrt{\frac{mh}{4\pi}}$$

(iii) $\lambda_{\text{D.B.}} = \sqrt{\frac{150}{6}}\mathring{A} = 5\mathring{A}$ **and** $\Delta x.\Delta p \geq \frac{h}{4\pi}; p = \frac{h}{\lambda}$ **or** $\Delta p = \frac{h}{\lambda^2}\Delta\lambda$

$\Rightarrow \quad \Delta x \frac{h}{\lambda^2} \times \Delta\lambda \geq \frac{h}{4\pi} \Rightarrow \frac{1}{\pi} \times \frac{10^{-9}}{\lambda^2} \times \Delta\lambda > \frac{1}{4\pi} \Rightarrow \Delta\lambda \geq \frac{2.5}{4} \times 10^{-10}$

Q.13 Answer:

$\sqrt{\frac{hm}{4\pi}}$

Solution:

(i) $\Delta x \Delta p = \frac{h}{4\pi} \Rightarrow \Delta p^2 = \frac{h}{4\pi}$

$$m^2\Delta v^2 = \frac{h}{4\pi} \Rightarrow \Delta v = \frac{1}{2\,m}\sqrt{\frac{h}{\pi}}$$

(ii) $\Delta x = \sqrt{\frac{h}{4\pi m}}, \Delta x \Delta p = \frac{h}{4\pi}$

$$\sqrt{\frac{h}{4\pi m}}\Delta p = \frac{h}{4\pi}; \Delta p = \sqrt{\frac{mh}{4\pi}}$$

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(iii) $\lambda_{\text{D.B.}} = \sqrt{\frac{150}{6}}\mathring{A} = 5\mathring{A}$ **and** $\Delta x.\Delta p \geq \frac{h}{4\pi}; p = \frac{h}{\lambda}$ **or** $\Delta p = \frac{h}{\lambda^2}\Delta\lambda$

$\Rightarrow \quad \Delta x \frac{h}{\lambda^2} \times \Delta\lambda \geq \frac{h}{4\pi} \Rightarrow \frac{1}{\pi} \times \frac{10^{-9}}{\lambda^2} \times \Delta\lambda > \frac{1}{4\pi} \Rightarrow \Delta\lambda \geq \frac{2.5}{4} \times 10^{-10}$

Q.14 Answer:

$0.625\mathring{A}$

Solution:

(i) $\Delta x \Delta p = \frac{h}{4\pi} \Rightarrow \Delta p^2 = \frac{h}{4\pi}$

$$m^2\Delta v^2 = \frac{h}{4\pi} \Rightarrow \Delta v = \frac{1}{2\,m}\sqrt{\frac{h}{\pi}}$$

(ii) $\Delta x = \sqrt{\frac{h}{4\pi m}}, \Delta x \Delta p = \frac{h}{4\pi}$

$$\sqrt{\frac{h}{4\pi m}}\Delta p = \frac{h}{4\pi}; \Delta p = \sqrt{\frac{mh}{4\pi}}$$

(iii) $\lambda_{\text{D.B.}} = \sqrt{\frac{150}{6}}\mathring{A} = 5\mathring{A}$ **and** $\Delta x.\Delta p \geq \frac{h}{4\pi}; p = \frac{h}{\lambda}$ **or** $\Delta p = \frac{h}{\lambda^2}\Delta\lambda$

$\Rightarrow \quad \Delta x \frac{h}{\lambda^2} \times \Delta\lambda \geq \frac{h}{4\pi} \Rightarrow \frac{1}{\pi} \times \frac{10^{-9}}{\lambda^2} \times \Delta\lambda > \frac{1}{4\pi} \Rightarrow \Delta\lambda \geq \frac{2.5}{4} \times 10^{-10}$

Q.15 Answer:

$7200\mathring{A}$

Solution:

Kinetic energy (K.E.) $= 4.55 \times 10^{-25}$ J

Mass of electron $(m_e) = 9.1 \times 10^{-31}$ kg

$h = 6.6 \times 10^{-34}$ J $-$ sec

de Broglie wavelength $(\lambda) = \dfrac{h}{\sqrt{2\,K.E m_e}}$

$= \dfrac{6.6 \times 10^{-34}\text{ J} - \text{sec}}{\sqrt{2 \times 4.55 \times 10^{-25} \times 9.1 \times 10^{-31}}}$

$= 0.72 \times 10^{-6}$ m

Q.16 Answer:

Directly proportional to its velocity

Solution:

$\lambda = \frac{h}{mv}; KE = \frac{1}{2}mv^2$

$$mv = \frac{2KE}{v} \quad \therefore \lambda = \frac{h}{\frac{2KE}{v}} = h\left(\frac{v}{2KE}\right)$$

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Q.17 Answer:

$\frac{1240}{2.55}$ nm

Solution:

Total nodes $= \mathrm{n}-1$

for initial orbit $= \mathrm{n}-1=2+1 \Rightarrow \mathrm{n}=4$

for final orbit $= n-1=1 \Rightarrow n=2$

$$\Delta \mathrm{E}=13.6 \mathrm{Z}^2\left(\frac{1}{\mathrm{n}_1^2}-\frac{1}{\mathrm{n}_2^2}\right) \mathrm{eV}=13.6 \times 1^2$$

$$\left(\frac{1}{2^2}-\frac{1}{4^2}\right)=2.55 \mathrm{eV}$$

$$\therefore \lambda=\frac{1240}{2.55} \mathrm{~nm}$$

Q.18 Answer:

A-p; B-p,q,s ; C- q,s ; D-q,s;

Solution:

((A) - p, (B) - pqs, (C) qs, (D) - qs)

(A) s-orbital $\because \mathrm{r}=0, \psi=0$ **and 3 radial nodes** $\Rightarrow 4 \mathrm{~s}$

(B) 3 radial nodes $(s, p, d) \Rightarrow 4s, 5_{px} 6d_{xy}$

(C) Angular probability is dependent of θ **and** ϕ **for** $5_{\mathrm{px}}, 6 \mathrm{~d}_{\mathrm{xy}}$ **(s orbital is spherically symmetric hence independent)**

(D) Atleast one angular node $\Rightarrow 5 \mathrm{p}_{\mathrm{x}}(1) ; 6 \mathrm{~d}_{\mathrm{xy}}(2)$

Q.19 Answer:

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Solution:

$$\Psi=\left(2 / \pi C^2\right)^{3 / 4} e^{-\left(x^2+y^2+z^2\right) / C^2}$$

$$|\Psi|^2 dx dy dz=\left(2 / \pi C^2\right) e^{-2\left(x^2+y^2+z^2\right) / c^2} dx dy dz$$

$$=\left[2 /\left(4 \pi n m^2\right)\right]^{3 / 2} e^{-2\left[(1.2)^2+(-1)^2+(0)^2\right] / 4}(0.004 \mathrm{~nm})^3$$

$$=1.15 \times 10^{-9}$$

Ans. $K=9$

Q.20 Answer:

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Solution:

Energy required per molecule in the following process may be given as

$$\mathrm{O}_2 \xrightarrow{\mathrm{hv}} \mathrm{O}+\mathrm{O}^*$$

$$\mathrm{E}=\frac{\mathrm{hc}}{\lambda}=\frac{6.626 \times 10^{-34} \times 3 \times 10^8}{174 \times 10^{-9}}=11.424 \times 10^{-19} \text{ Jmolecule}^{-1}$$

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Energy for, $O \rightarrow O^*$ **is** 3.15×10^{-19} J

Energy for $O_2 \rightarrow 2O$ **will be** $= 11.424 \times 10^{-19} - 3.15 \times 10^{-19}$

$= 8.274 \times 10^{-19} \times 6.023 \times 10^{23} \times 10^{-3}$ kJ mole^{-1} $= 498.3$ kJ mol^{-1}

$x = 1$

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